Orbit Guidance Theory

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Vector Field Guidance for Circular Loiter

Local Coordinate Frame

The geographic space is locally flat. We introduce scale factors based on current vehicle position (λ, φ):

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, \qquad M(\varphi) = \frac{a(1 - e^2)}{\left(1 - e^2 \sin^2 \varphi\right)^{3/2}}$$

where $e^2 = f(2 - f)$, $f = 1/298.257223563$, $a = 6378137$ m.
Define local planar coordinates with scaling:

$$x = N(\varphi) \cos \varphi \cdot \lambda, \qquad y = M(\varphi) \cdot \varphi$$

The dynamics in this local frame:

$$\dot{x} = v \sin \psi, \qquad \dot{y} = v \cos \psi$$

This is a planar system with constant speed v and free heading ψ.

Problem Setup

Given:

- Orbit center in geographic coordinates: (λ_c, φ_c)
- Desired orbit radius: R (meters)
- Current vehicle position: (λ, φ)

Convert to local planar frame:

$$x_c = N(\varphi) \cos \varphi \cdot \lambda_c, \qquad y_c = M(\varphi) \cdot \varphi_c$$

$$x = N(\varphi) \cos \varphi \cdot \lambda, \qquad y = M(\varphi) \cdot \varphi$$

Relative position to orbit center:

$$\Delta x = x - x_c, \qquad \Delta y = y - y_c$$

Polar coordinates:

$$r = \sqrt{\Delta x^2 + \Delta y^2}, \qquad \theta = \operatorname{atan2}(\Delta x, \Delta y)$$

Radial error:

$$e_r = r - R$$

Vector Field Guidance Law

The heading command is:

$$\boxed{\psi = \theta + \lambda \cdot \left[\operatorname{atan}\left(\frac{R - r}{k}\right) + \frac{\pi}{2}\right]}$$

where:

- $k > 0$ = transition gain (meters)
- $\lambda = +1$ (CCW) or $\lambda = -1$ (CW)

Behavior

When $r = R$ (on orbit):

$$\psi = \theta + \lambda \cdot \frac{\pi}{2}$$

Pure tangential motion.

When $r > R$ (outside):

$$\frac{R - r}{k} < 0 \quad \Rightarrow \quad \text{heading turns inward}$$

When $r < R$ (inside):

$$\frac{R - r}{k} > 0 \quad \Rightarrow \quad \text{heading turns outward}$$

Geographic Rate Equations

Apply computed ψ to geographic dynamics:

$$\dot{\lambda} = \frac{v \sin\psi}{N(\varphi)\cos\varphi}, \qquad \dot{\varphi} = \frac{v \cos\psi}{M(\varphi)}$$

Implementation

At each step:

1. Compute scale factors $N(\varphi)$, $M(\varphi)$ at current φ
2. Convert (λ, φ) and (λ_c, φ_c) to local (x, y) and (x_c, y_c)
3. Compute Δx, Δy, then r and θ
4. Compute heading ψ from vector field
5. Integrate $\dot{\lambda}$ and $\dot{\varphi}$

The scale factors update continuously as the vehicle moves, but for small regions this variation is negligible.